AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 27 · STUDENT LESSON

Verify Important Outputs

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

Check each claim. Keep the units. Follow the evidence.

Audit sources, verify a ribbon calculation in two ways and distinguish an error from an unresolved claim.

Keep one habit: Split the draft → choose each check → report the supported correction and its limits.

MAIN LESSON · SECTION 1

Which claims can the evidence actually support?

> Separate facts, calculations and unresolved promises.

Start with a mixed-quality answer

One paragraph can contain a correct count, a wrong unit and an unsupported promise. Check each claim separately instead of giving the whole paragraph one yes-or-no verdict.

Recall format versus factual support

A response follows the requested four-line format but invents a missing start time. Has it passed the task? Explain.

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Your learning goals

- I can separate an answer into **claim**s and connect each **claim** to relevant **evidence**.
- I can recalculate quantities with consistent **unit**s and a reasonableness check.
- I can report supported, contradicted and unresolved **claim**s without inventing missing **evidence**.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Prompts Are Design Instructions. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Privacy by Choosing Less. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Good Data, Bad Data. Give an example and a mistake that the idea helps you avoid.

Fluent AI **output** is not **evidence**; important **claim**s need **independent** support.



LEARNING TARGET A generated report says 18 of 24 tests passed, so success was **80%**. Verify the **claim** in two ways and write a corrected sentence.

Learner name:

Readiness for the connected explanation

This week’s end goal

By the end, I can verify quantities with consistent **unit**s and separate arithmetic from unsupported **claim**s.

MAIN LESSON · SECTION 2

Meet materials card R and the draft

> Keep the current ribbon activity separate from old card Q.

Separate the paragraph into claims

An answer can contain a mixture of correct, incorrect and unverified statements. Checking one sentence does not validate every other sentence. First separate the answer into individual **claim**s.

Then ask what **evidence** would support each one: a record, a measurement, a calculation or some combination.

Read the current record and model assumptions

Our fictional ribbon activity uses current materials card R. R states: four rolls of ribbon, each exactly 2 metres long; twelve pieces will each use 50 centimetres; no cutting loss is included in this paper model.

One metre equals 100 centimetres. An old card Q describes a different activity with three rolls. These are provided classroom records, not instructions to cut real materials.

Inspect the five draft claims

A draft says “R gives four rolls. The starting length is 8 metres. The pieces use 600 metres. Two metres remain. Every participant will enjoy the activity.” Do not approve or reject the whole paragraph at once.

Inspect the roll count, starting length, used length, remaining length and enjoyment **claim** separately.

▼ Earlier wording, alternatives and picture transcripts

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



MYSTERY A generated paragraph confidently **claim**s that an animal lives in a region; the wording is not **evidence** that the **claim** is true.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first **prediction** and state the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

A closer explanation

Match each claim to a suitable check

> A nearby citation supports only what its actual contents establish.

Inspect the claim-by-claim evidence table

Different claims need different checks

CLAIM	EVIDENCE PATH	DECISION
Four rolls	Read current card R	Supported
8 metres at start	4 rolls × 2 metres	Supported
600 metres used	12 × 50 cm = 600 cm	Wrong unit
Everyone will enjoy it	No enjoyment evidence	Unresolved

A correct source reference is a starting point. Read what the record actually says and check every derived quantity.

Check that a citation is available and relevant

A **citation** is useful only if you can locate the **source** and confirm that it supports the specific **claim**. The title “Materials card R” beside a paragraph does not make the enjoyment **claim** part of R.

If a cited card Z is not supplied or cannot be found, that **citation** cannot currently verify its **claim**. Record the gap rather than inventing Z’s contents.

Check the activity and version

- Check whether the **source** covers the right activity, version, population or time. Q's three rolls may be accurate for its own activity but irrelevant to the current four-roll task.

  In real work, also inspect who created the record, how they obtained the information and whether there is a correction or better **source**.

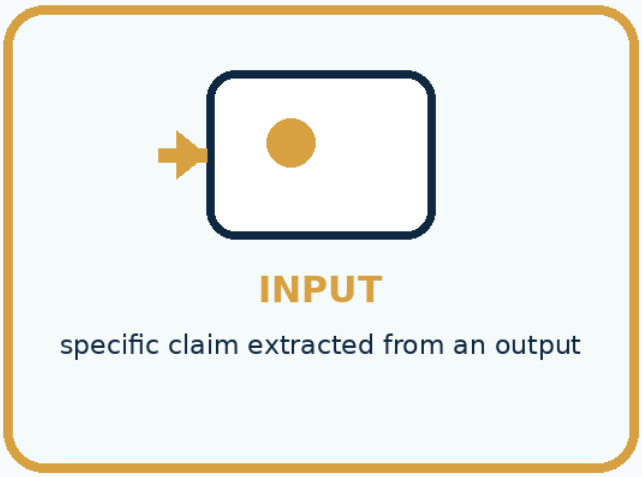
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

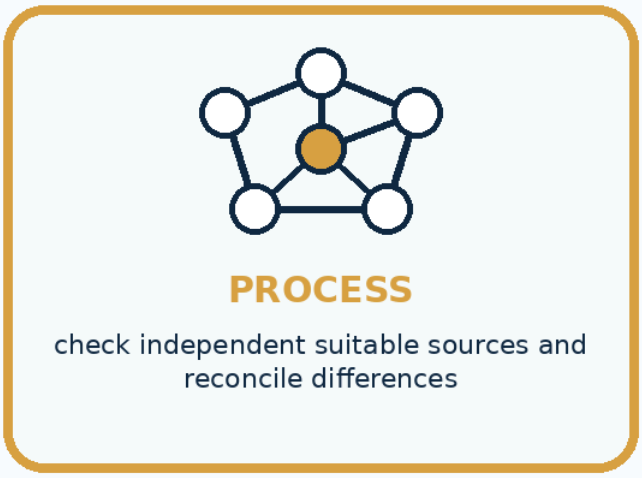
▼ Earlier wording, alternatives and picture transcripts

VISUAL **MECHANISM**

Input → Process → Output

Follow the information instead of imagining magic inside the machine.







INPUT specific **claim** extracted from an **output**

PROCESS check **independent** suitable **source**s and reconcile differences

OUTPUT supported, corrected, uncertain, or removed **claim**

CORE IDEA Fluent AI **output** is not **evidence**; important **claim**s need **independent** support.

UNDER THE HOOD **Verification** is **claim**-level **evidence** work. **Source** independence, authority, direct support, recency, and agreement matter. Multiple pages repeating the same upstream error are not **independent** confirmation.

Draw a fresh example with all three stages:

A closer explanation

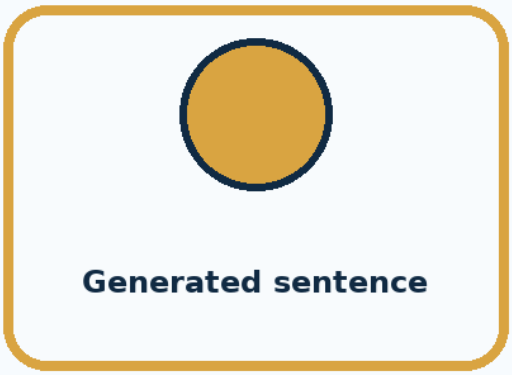
Connect the picture and the explanation

MAIN LESSON · SECTION 4

Compare evidence with repeated assertions

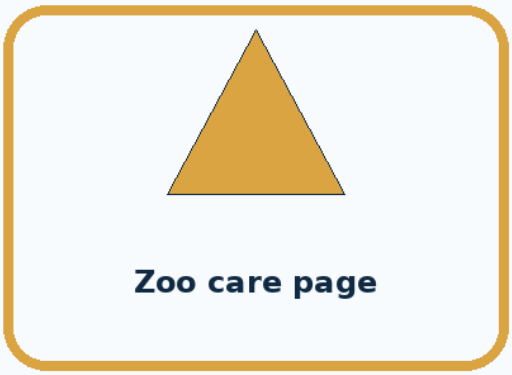
> Two copied pages are not two independent confirmations.

Pictures paired with their evidence



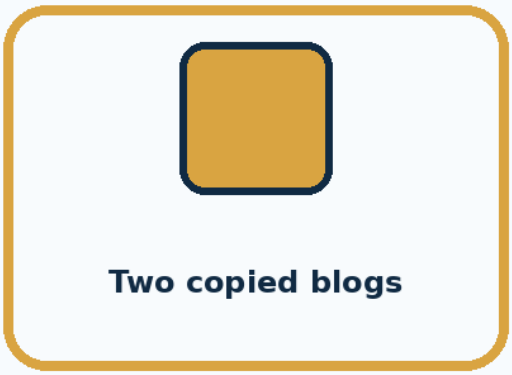
The case to investigate

CASE 1 — Generated sentence Clue: fluent **output**. Meaning: **claim**, not **evidence**.



The case to investigate

CASE 2 — Zoo care page Clue: direct animal information. Meaning: possible **source**.



The case to investigate

CASE 3 — Two copied blogs Clue: same origin. Meaning: not **independent**.



The case to investigate

CASE 4 — Conflicting references Clue: **evidence** disagrees. Meaning: report uncertainty.

Discuss the confusing pair

- Which two pictures are easiest to confuse? What **evidence** separates them?

Keep the limit in view

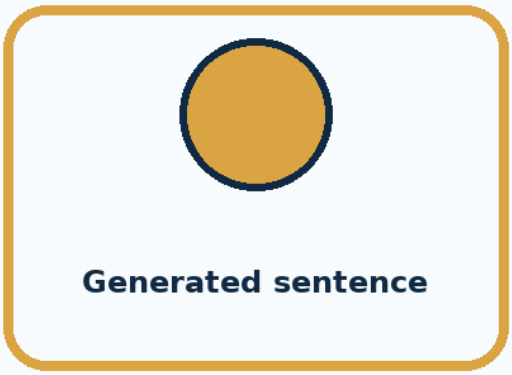
BOUNDARY Two search results are not two **independent source**s when both copy the same page.

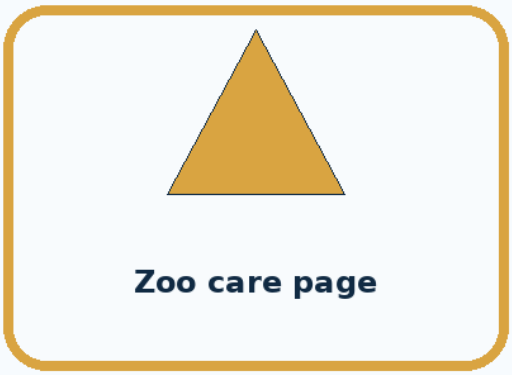
▼ Earlier wording, alternatives and picture transcripts

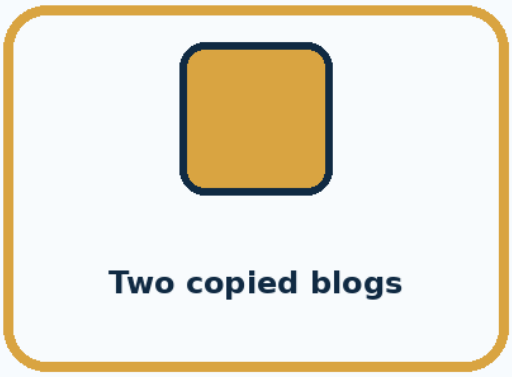
PICTURE GALLERY

Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.









CASE 1 — Generated sentence Clue: fluent **output**. Meaning: **claim**, not **evidence**.

CASE 2 — Zoo care page Clue: direct animal information. Meaning: possible **source**.

CASE 3 — Two copied blogs Clue: same origin. Meaning: not **independent**.

CASE 4 — Conflicting references Clue: **evidence** disagrees. Meaning: report uncertainty.

Change one thing in the pictured case

Change: Make ten bookmarks instead of eight.

Prediction: 100 cm, or 1 m, remains.

Reason: 600−10×50=100 cm.

Trace the separate bookmark calculation

> These four 150 cm rolls are a different dataset from card R's four 2 m rolls.

The bookmark facts and explicit assumptions

A fictional craft sheet calculates ribbon left after making bookmarks.

Four rolls each contain exactly 150 cm. Eight bookmarks use exactly 50 cm each.

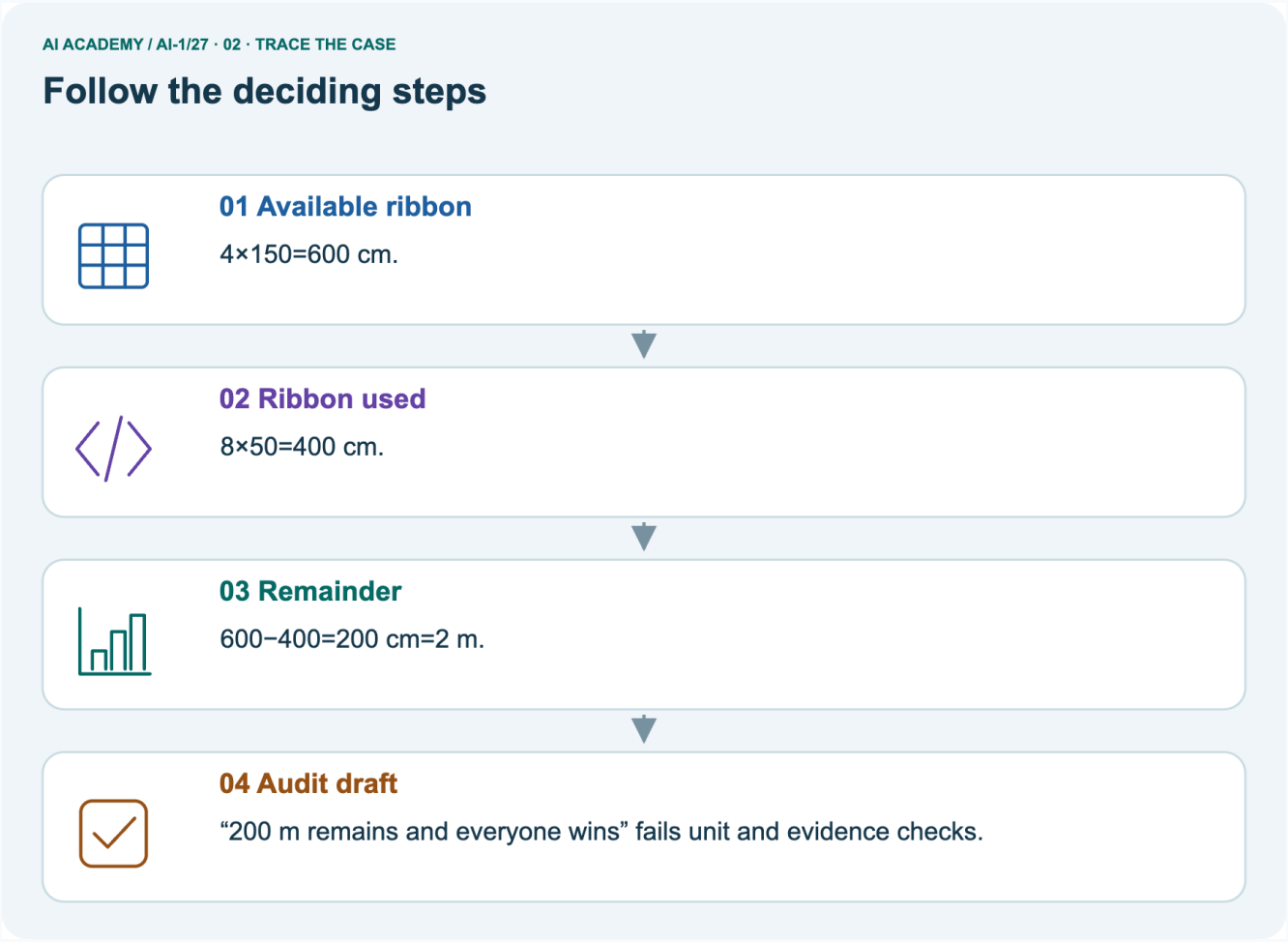
No other ribbon loss is assumed; no competition result is recorded.

Inspect the labelled quantity comparison



Follow the labelled picture with the matching explanation. Open the vector diagram

Trace the bookmark arithmetic and claim audit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Available ribbon	4×150=600 cm.
2. Ribbon used	8×50=400 cm.
3. Remainder	600−400=200 cm=2 m.
4. Audit draft	“200 m remains and everyone wins” fails unit and evidence checks.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Explain units and unsupported claims

- All arithmetic first uses centimetres. Converting only after subtraction avoids mixing **unit**s. The record supports available, used and remaining lengths under its no-loss assumption. It does not support a promise about winning.

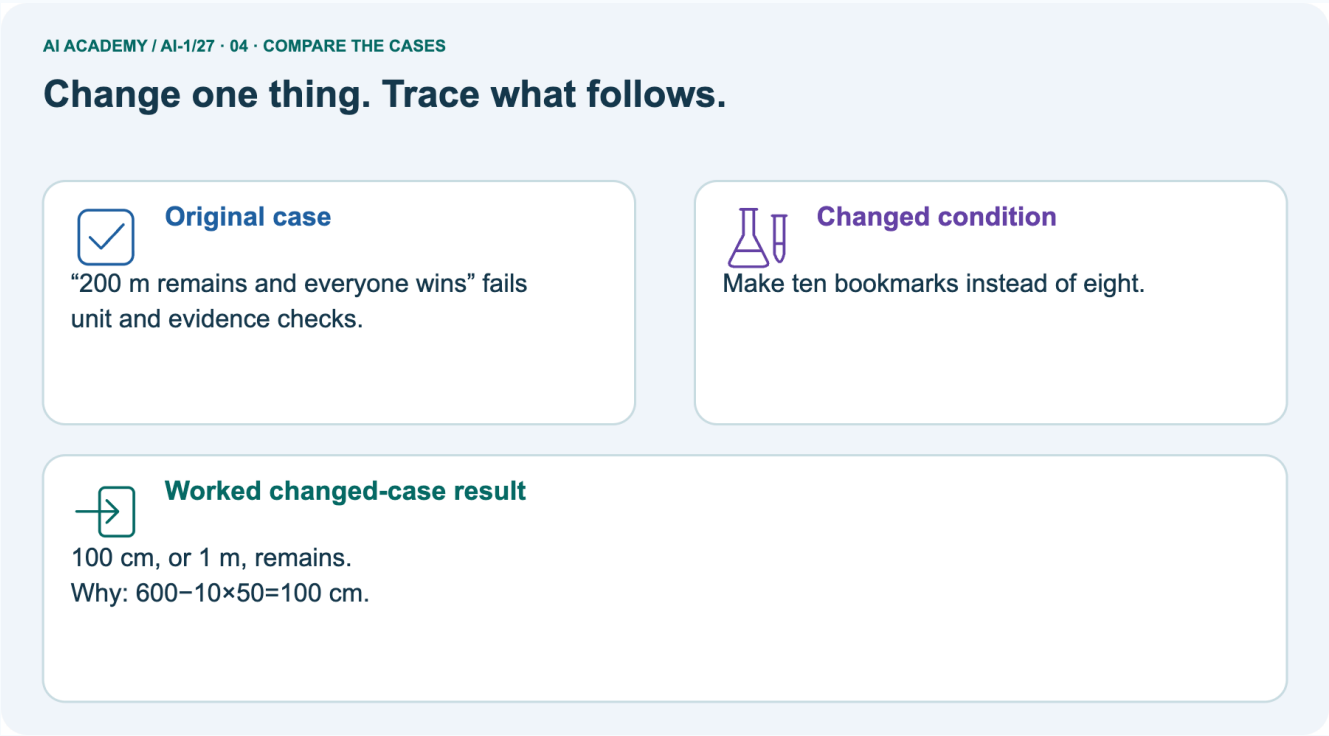
  A numerically correct total elsewhere in a draft cannot rescue a wrong **unit** or an invented **claim**.

- Read the labelled picture: Correct the draft and identify which part needs another **source** rather than a calculation.

Predict ten bookmarks

- Before opening the changed-bookmark picture, make ten bookmarks from the same four 150 cm rolls, still assuming no other loss. Predict the remaining length in centimetres and metres. Which promise about winning remains unsupported?

▼ Check the changed calculation



Follow the labelled picture with the matching explanation. Open the vector diagram

Keep the no-loss assumption visible

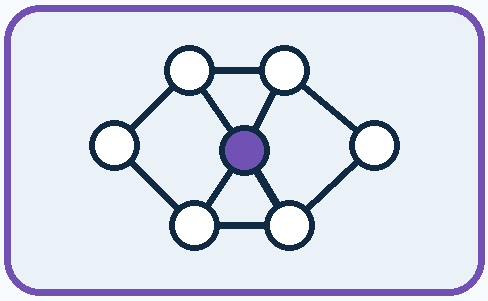
Real cutting losses or inaccurate roll lengths would change this toy calculation; the assumption must be declared.

▼ Earlier wording, alternatives and picture transcripts

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a **claim** to observable **evidence**.



CASE A generated paragraph confidently **claim**s that an animal lives in a region; the wording is not **evidence** that the **claim** is true.

STEP 1 — OBSERVE specific **claim** extracted from an **output**

STEP 2 — TRACE check **independent** suitable **source**s and reconcile differences

STEP 3 — CONCLUDE supported, corrected, uncertain, or removed **claim**

MODEL REASONING Check the exact **claim**, not the whole paragraph. Use **source**s with direct relevant **evidence**, note publication context, and explain whether they **independent**ly agree.

If **evidence** is weak or conflicting, revise or remove the **claim**.

DO NOT OVER**CLAIM** Two search results are not two **independent source**s when both copy the same page.

What new **evidence** could change this answer?

A closer explanation

Quantity	Calculation in centimetres	Equivalent in metres
Starting length	4 × 200 cm = 800 cm	8 m

Quantity	Calculation in centimetres	Equivalent in metres
Length used	12 × 50 cm = 600 cm	6 m
Length remaining	800 cm − 600 cm = 200 cm	2 m

Convert to one length **unit** before subtracting. Two metres is 200 centimetres, so four rolls give 800 centimetres. The twelve pieces use 600 centimetres, leaving 200 centimetres, or 2 metres.

Subtracting the bare numbers 8 − 600 would mix metres with centimetres and would not describe the remaining length correctly.

Use a second route as a check: 50 centimetres is half a metre, so twelve pieces use six metres. Eight minus six leaves two metres.

A reasonableness check also asks whether the used length is at most the starting length and whether starting length equals used plus remaining.

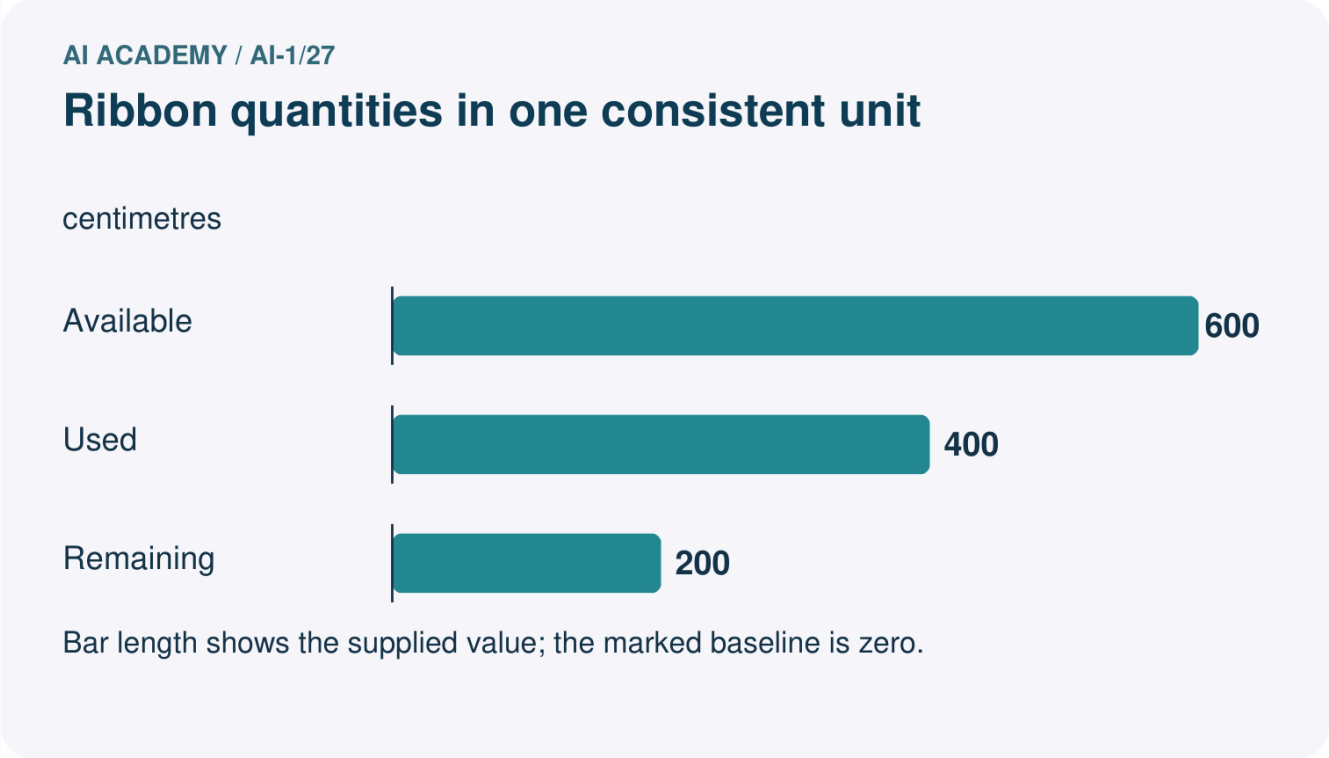
These checks can reveal mistakes; they do not replace checking the original **input**s and assumptions.

Learn from the worked case

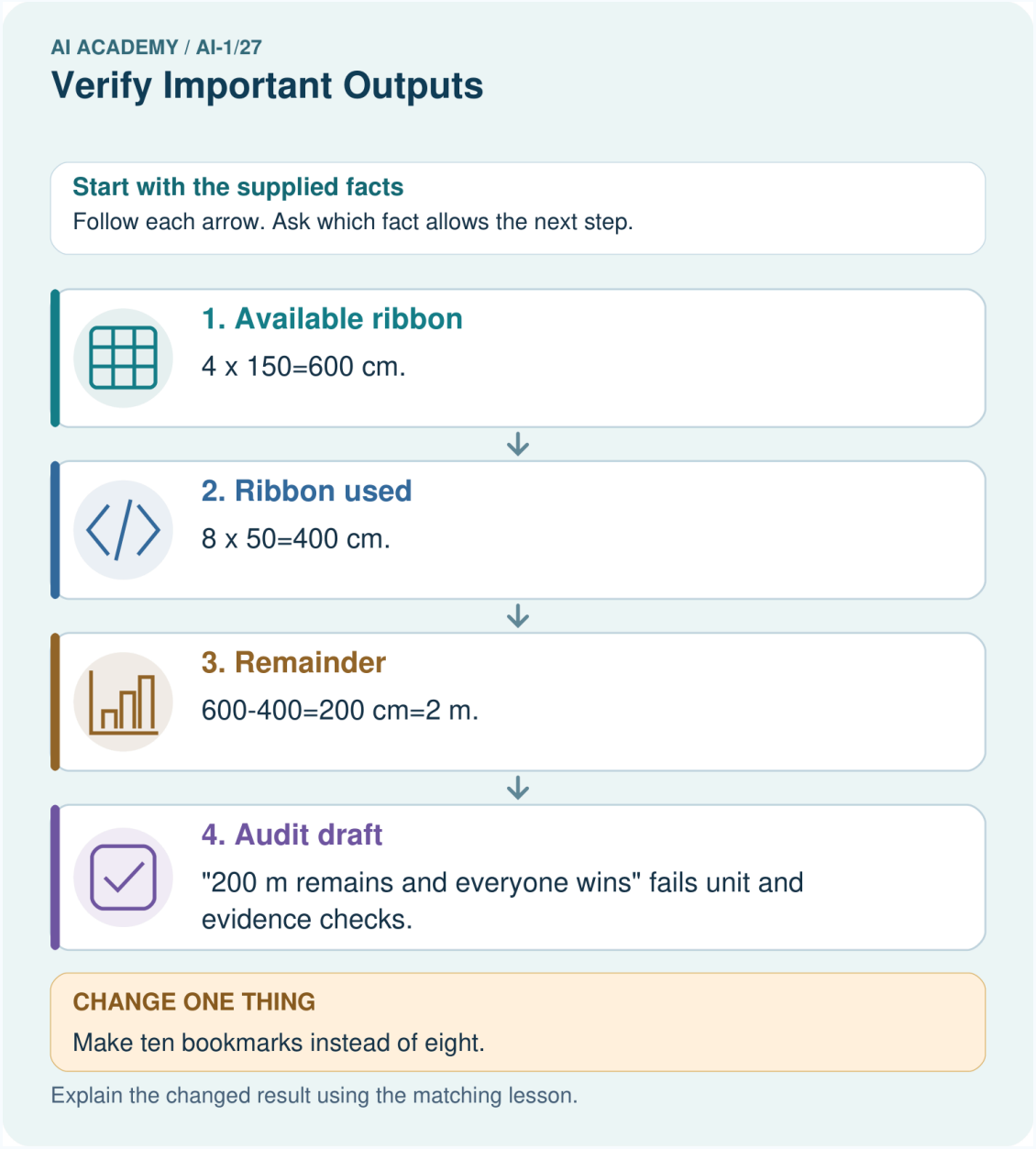
Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Verify Important Outputs

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing: Make ten bookmarks instead of eight. Predict the result before checking the explanation. What remains the same?

Write a correction that preserves its conditions

> Name the source, expression, unit and unresolved evidence.

Model a complete correction using card R

A useful correction names the original **claim**, relevant **source**, calculation with **unit**s and revised conclusion. For example: R lists four 2-metre rolls and twelve 50-centimetre pieces.

The pieces use 600 centimetres, or 6 metres; the paper model leaves 2 metres when no cutting loss is included. This preserves the condition on which the result depends.

Explain what would change the model

If the real activity loses ribbon at each cut, the no-loss result cannot simply be reused. Obtain that missing information or label the answer as a model with an explicit assumption.

Similarly, if you cannot access a **source**, mark the **claim** unresolved and identify what **evidence** would settle it. Repeating a confident answer is not a new measurement.

Repair both tempting claims

- **CLAIM** A Professional wording is **evidence** of accuracy.
- **CLAIM** B The first two agreeing links settle every **claim**.

STRONG ANSWER FRAME I **claim** _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

▼ **Earlier wording, alternatives and picture transcripts**

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case	Evidence	What it shows
Generated sentence	fluent output	claim, not evidence
Zoo care page	direct animal information	possible source
Two copied blogs	same origin	not independent
Conflicting references	evidence disagrees	report uncertainty

Repair two tempting claims

Repair:

Repair:

A closer explanation

> **In a changed paper model, each of the twelve pieces uses 50 cm and an extra 2 cm is lost for each piece. Calculate the total loss and the remaining ribbon from the same four rolls. State which assumption changed**

MAIN LESSON · SECTION 8

Practise: distinguish wrong from not yet verified

> Lack of enjoyment evidence does not establish that nobody will enjoy the activity.

Classify the unit error and enjoyment claim

The **claim** “600 metres used” conflicts with the stated **input**s and calculation. The **claim** “everyone will enjoy it” is not supported by R or by any supplied enjoyment **evidence**. That does not establish that nobody will enjoy it.

State “not verified from these records” rather than changing an unsupported positive **claim** into an unsupported negative **claim**.

Follow repeated claims back to their origin

Suppose three posters repeat the same number and each copied it from one unverified draft. That is three copies of one **claim**, not three **independent** measurements. Follow references back to their origin.

A separate check of R and the arithmetic provides more useful **evidence** than counting copies that repeat the same mistake.

Repair the unit error in W2

Use workbook W2 for the connected example “Verifying **claim**s and quantities”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Challenge the citation claim

- Use the worked case for Verifying **claim**s and quantities. Proposed explanation: “A **citation** proves every nearby **claim**”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Return to the bookmark draft

Correct the draft and identify which part needs another **source** rather than a calculation.

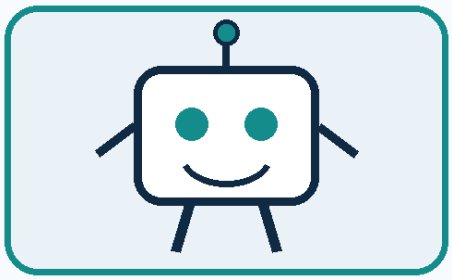
This is supported practice. Keep the separately named **independent** case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed **independent**ly later.



Explain how the picture story demonstrates this idea: Fluent AI **output** is not **evidence**; important **claim**s need **independent** support.

Use a fresh example to explain **claim**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: The first two agreeing links settle every **claim**.

Invent one edge case and name the safe response.

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

> **Audit the draft from section one. For each of its five claims, state supported, contradicted or unresolved under the supplied evidence. Correct the used-length claim and explain why the correct remaining-length claim still needs its own calculation**

Practise with less help: Repair a unit error

Check your reasoning

Explain the picture

> **My explanation using a pictured step and the given facts**

MAIN LESSON · SECTION 9

Investigate source independence with supplied records

> Use a fictional animal so the evidence and its origins are fully inspectable.

The draft and three source cards

Fictional source audit; no animal research or internet search is needed. Draft: “The invented silver kitebird was observed in Meadow Park on Monday, and silver kitebirds are never observed anywhere else.” Source A is a dated observer log recording one silver kitebird at Meadow Park on Monday. Source B is a poster explicitly copied from A. Source C is a

separately made observer log recording one silver kitebird at River Park on Tuesday. Treat the supplied logs as the accepted observations for this paper exercise; they are not real wildlife records.

Predict, audit and correct each claim

1. Predict which parts of the draft each source can support. Make separate rows for the Monday sighting and the claim that silver kitebirds are never observed outside Meadow Park. Record each source's origin and direct evidence. Explain why B adds no independent observation, how C bears on the "never observed elsewhere" claim, and why these logs do not establish every place kitebirds can be found. Write a corrected statement limited to the supplied observations. Mark any actual field investigation "not run".

2. Keep a normal case and an edge case clearly separated.

3. Record what happened—even when it does not match your **prediction**.

4. Explain the result, one limitation, and the next useful test.

Keep observations and remaining limits

Case	Prediction	Observed result	What it means
Normal			
Edge			

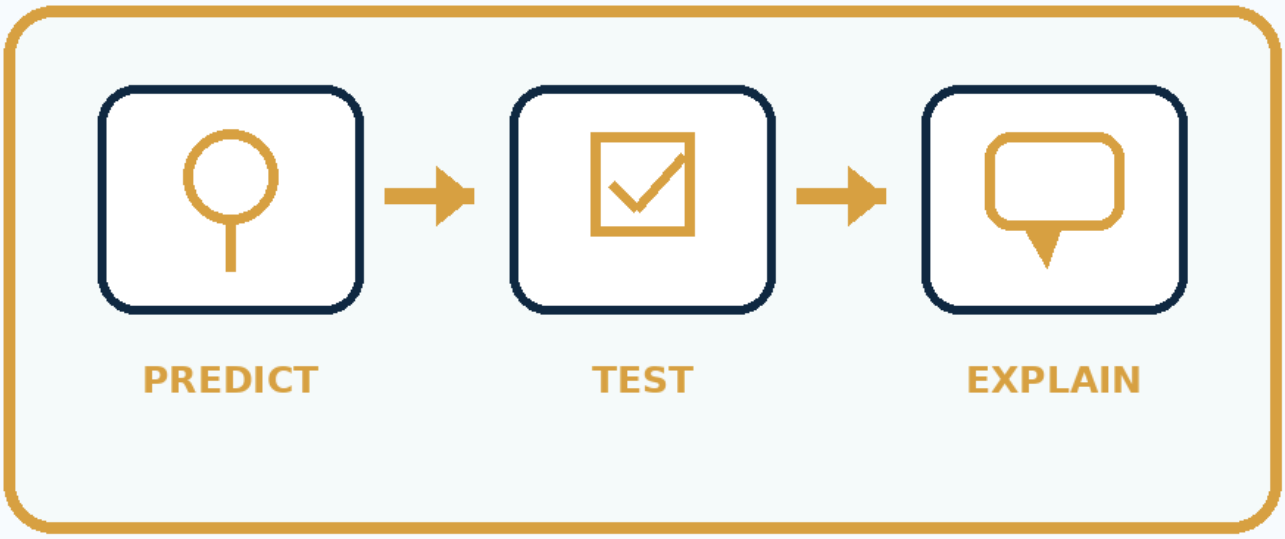
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep "not run" if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

EXPLORER LAB

Two-Source Claim Check

Predict first, test honestly, and explain what the **evidence** means.



State your **prediction** before the result is known.

Complete the task: Use a two-**source** checklist on a generated animal fact.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: audit the new mass-unit draft

> Use craft card M before consulting feedback.

The independent record and four claims

Fictional craft card M states five sealed bags of modelling material at exactly 250 g per bag. Three models each use 300 g, with no other loss.

A draft says “M supplies 1250 g; the models use 900 g; 350 kg remains; every model will win a prize.” Check all four **claim**s, show the remaining mass in grams, and explain the status of the prize **claim**. Use 1 kg = 1000 g.

Keep your first attempt

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the **independent** answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the **independent** case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Audit a new model with mass units

> **My independent explanation, trace and limits**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Week 27 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

An assistant says the school library opens at 8, but today's official notice says 9. Which **source** should guide today's visit? Explain the check.

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Require human confirmation for important sorter **output**s and attach the **evidence** record used to approve each rule or label.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Transfer practice from the original programme

A generated report says 18 of 24 tests passed, so success was **80%**. Verify the **claim** in two ways and write a corrected sentence.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Attach the evidence behind important sorter outputs

> Verify a particular claim rather than approving an entire explanation.

This week’s project decision

THIS WEEK'S CHECKPOINT Require human confirmation for important sorter **output**s and attach the **evidence** record used to approve each rule or label.

Check sorter calculations with explicit **unit**s and verify explanation **claim**s against their actual catalogue **evidence**.

Plan source and calculation checks

- What real classroom need does this serve?
- What information is necessary—and what should stay private?
- What happens inside the **process**?
- What normal and edge cases will you test?

- When should it say not sure or ask a person?

Inspect independence before counting sources

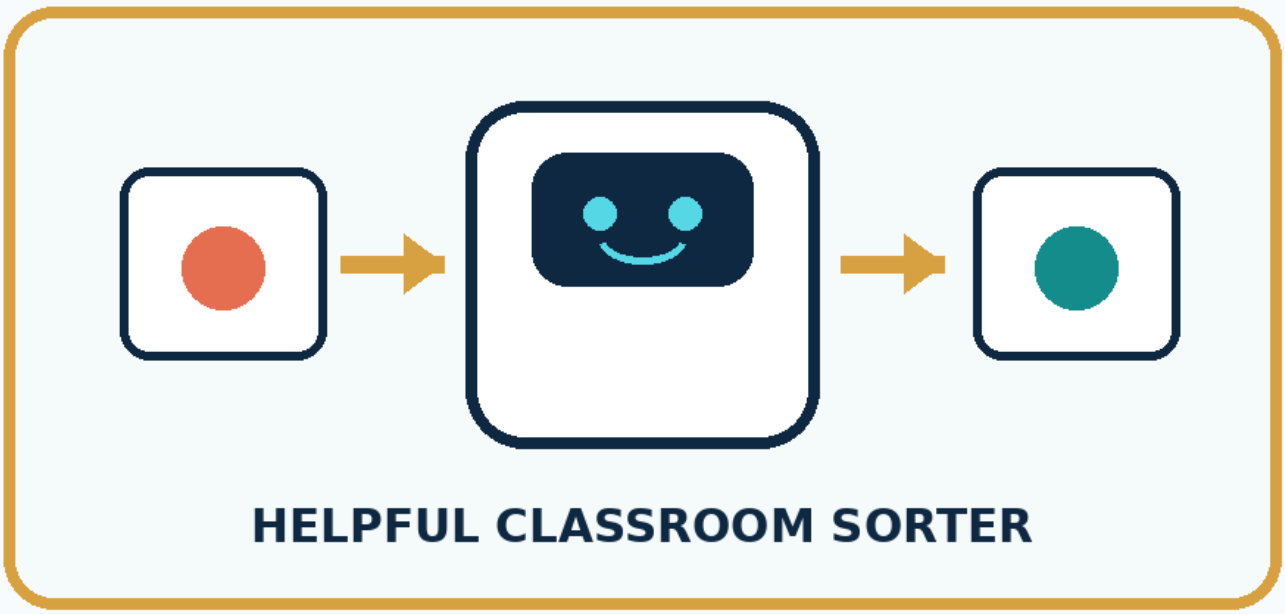
RESPONSIBLE DESIGN Two search results are not two **independent source**s when both copy the same page.

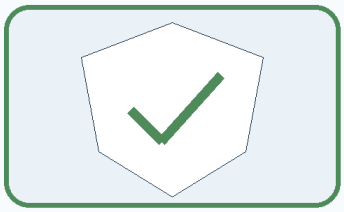
▼ **Earlier wording, alternatives and picture transcripts**

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.





My project decision:

Evidence for your project

MAIN LESSON · SECTION 16

Finish: justify each verification decision

> Close your notes and distinguish contradicted claims from unresolved ones.

Your exit explanation

- Close your notes. Explain this goal in your own words: Separate an answer into **claim**s and connect each **claim** to relevant **evidence**

> **My deciding step, evidence and one remaining question**

Choose the next changed-case check

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	Meaning	My example
claim	statement that can be checked	
source	origin of evidence	
independent	not merely copying the same origin	
corroborate	support a claim with separate evidence	

Terms for the connected explanation

Word	Meaning
Verification	Checking whether a claim is supported by appropriate evidence
Citation	A reference that helps a reader locate a source
Unit	The agreed measurement scale attached to a quantity, such as centimetres

Word	Meaning
Corroboration	Additional evidence that supports a claim, with its origin checked

Repair a claim using a word

- **CLAIM** TO REPAIR Professional wording is **evidence** of accuracy.
- Use one vocabulary word and one piece of **evidence** to repair it:

▼ Earlier wording, alternatives and picture transcripts

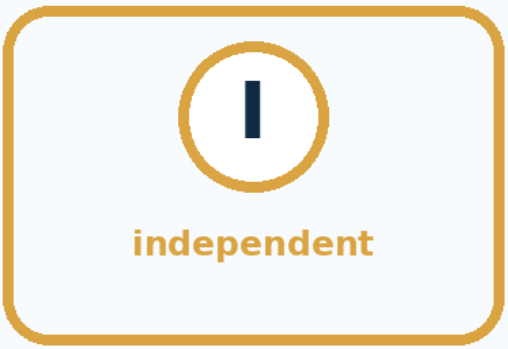
VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.









Misconception detective

WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

▼ Choose a stronger thinking challenge

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

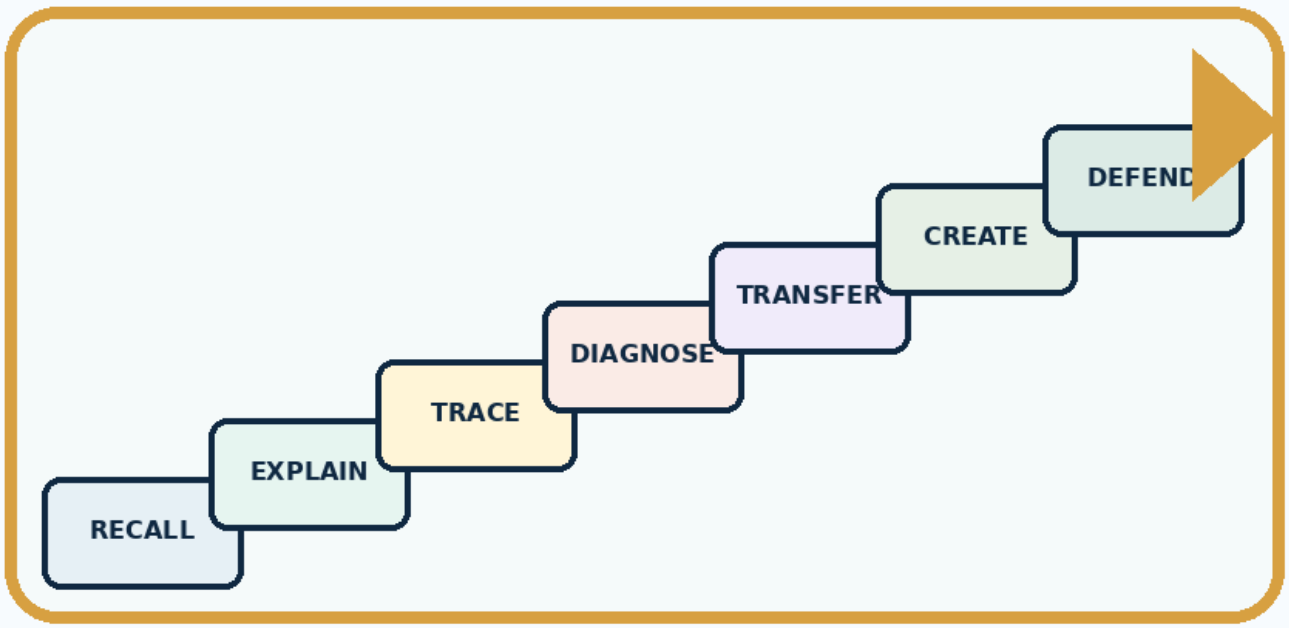
- RECALL Define **claim**.
- EXPLAIN Fluent AI **output** is not **evidence**; important **claim**s need **independent** support.
- TRACE Show the **input**, **process**, and **output**.
- DIAGNOSE Repair: Professional wording is **evidence** of accuracy.
- TRANSFER Apply the idea to a new home or classroom case.
- CREATE Design one revealing test.
- DEFEND Give a **claim**, **evidence**, **mechanism**, limit, and safe action.

▼ Earlier wording, alternatives and picture transcripts

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RETRY RULE A coached answer becomes mastery only after you solve a different case **independent**ly.

▼ **Reference: check your explanation**

REFERENCE · SECTION 12

Reference: check your explanation

> Use after practice. Audit a new model with mass units is the selected independent case.

Check the evidence and its limit

- ☐ Name or draw the fresh case.
- ☐ Trace **input**, **process**, and **output**.

- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Record support

My level: ☐ Not yet ☐ With one hint ☐ **Independent**ly ☐ I can teach it

▼ **Earlier wording, alternatives and picture transcripts**

MASTERY

Checklist for explaining this week's idea

Use after practice. Audit a new model with mass units is the selected independent case.



EXPLANATION CHECK Check each **claim** against its relevant record, calculate with consistent units and retain unresolved assumptions.

Independent evidence:

SELF-CHECK Check the exact **claim**, not the whole paragraph. Use **source**s with direct relevant **evidence**, note publication context, and explain whether they **independent**ly agree.

If **evidence** is weak or conflicting, revise or remove the **claim**.

NEXT BRIDGE Carry today's **evidence** habit into: Humans Set Boundaries.

▼ **Extra practice: Six groups of four make twenty-four**

REFERENCE · SECTION 13

Extra practice: Six groups of four make twenty-four

> Try the question before opening its worked explanation.

Read the supplied case and question

An answer says six groups of four make 28.

Verify **independent**ly.

Your explanation

- Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

▼ Check after trying

Six times four is 24. Draw six groups or add four six times. Confident wording does not change arithmetic.

Evidence or stage	Value or result
Group 1	4
Group 2	4
Group 3	4
Group 4	4
Group 5	4
Group 6	4

▼ Earlier wording, alternatives and picture transcripts

Week 27 Worked case

Use the supplied details to practise the weekly concept before attempting the **independent** question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Six groups of four make twenty-four

An answer says six groups of four make 28.

Verify **independent**ly.

Six times four is 24. Draw six groups or add four six times. Confident wording does not change arithmetic.

▼ **Reference: original lesson checklist**

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Separate an answer into claims and connect each claim to relevant evidence	Explain the deciding step, show a trace or test, and state one limit.
Recalculate quantities with consistent units and a reasonableness check	Explain the deciding step, show a trace or test, and state one limit.
Report supported, contradicted and unresolved claims without inventing missing evidence	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try **independent**ly → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Earlier wording, alternatives and picture transcripts**

Start with this question: A fictional craft sheet calculates ribbon left after making bookmarks.

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional craft sheet calculates ribbon left after making bookmarks.
- Which labelled step matches this detail? “200 m remains and everyone wins” fails **unit** and **evidence** checks.
- 1. Audit draft
- 2. Remainder
- 3. Ribbon used
- 4. Available ribbon
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Audit draft “200 m remains and everyone wins” fails **unit** and **evidence** checks. All arithmetic first uses centimetres.

Converting only after subtraction avoids mixing **unit**s. The record supports available, used and remaining lengths under its no-loss assumption. It does not support a promise about winning.

A numerically correct total elsewhere in a draft cannot rescue a wrong **unit** or an invented **claim**.

Changed-case question

- Changed-case practice: Make ten bookmarks instead of eight. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. 100 cm, or 1 m, remains.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: 100 cm, or 1 m, remains. 600−10×50=100 cm.

Repair and save your work

Review Prompts Are Design Instructions (ai-1/26). Goal: Write a prompt with a clear task, supplied **evidence** and checkable **output** requirements.

Short practice: What must happen if B instead conflicts with another equally authoritative duration record?. Return to this week and explain what you changed.

Review Privacy by Choosing Less (ai-1/24). Goal: Justify which information a stated task needs and remove unnecessary fields.

Short practice: For the original task, does adding an anonymous-looking pupil code improve the repair count?. Return to this week and explain what you changed.

Review Good Data, Bad Data (ai-1/21). Goal: Check records against a stated format, **unit** and meaning before using them. Short practice: Which repairs are **evidence**-backed in the original case, and which row must remain unknown?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved **independent** assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

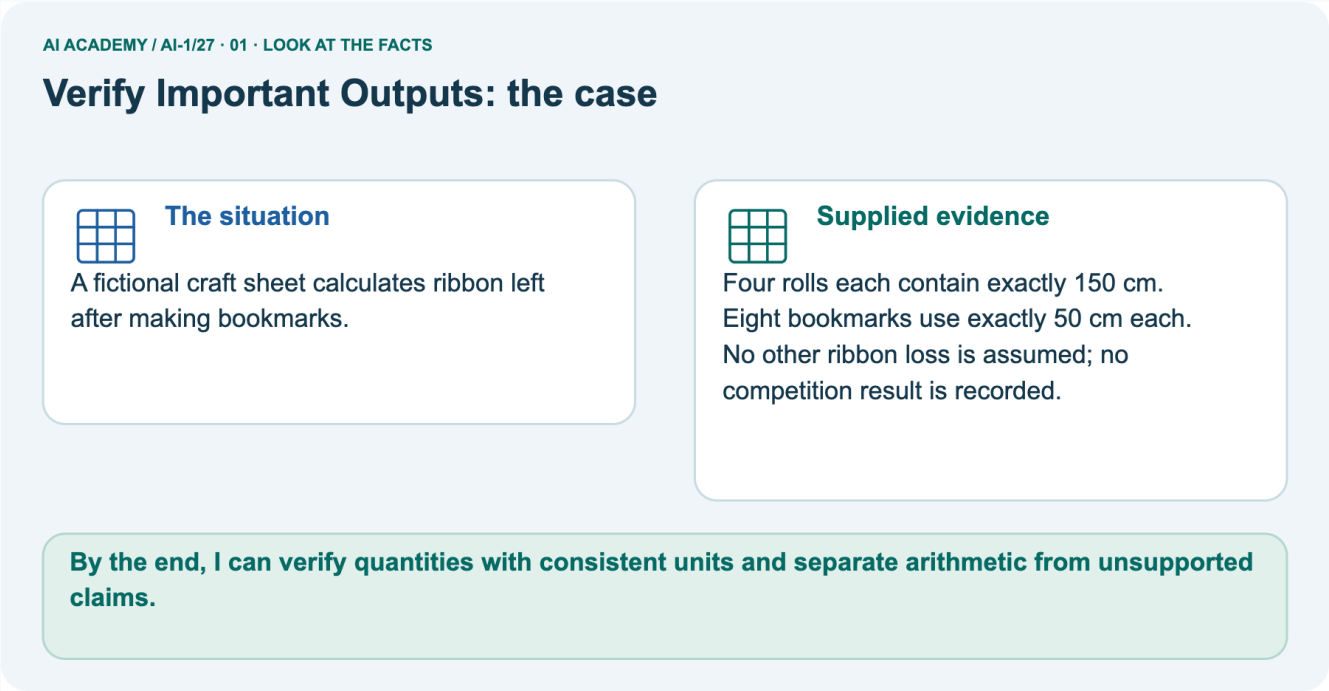
▼ Reference: the six-picture journey

REFERENCE · SECTION 18

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Verify Important Outputs: the case



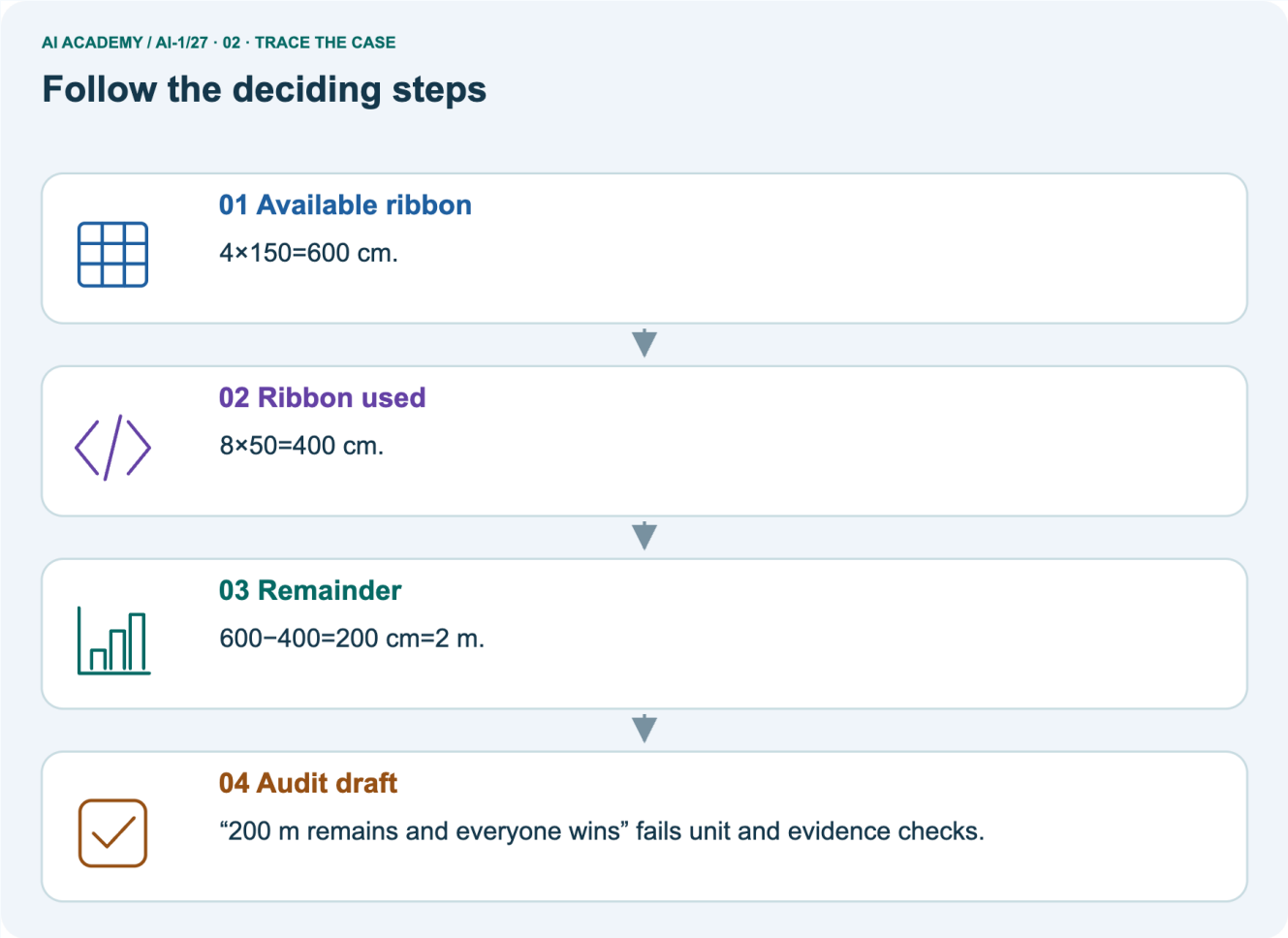
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional craft sheet calculates ribbon left after making bookmarks. Supplied facts: Four rolls each contain exactly 150 cm. Eight bookmarks use exactly 50 cm each.

No other ribbon loss is assumed; no competition result is recorded.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

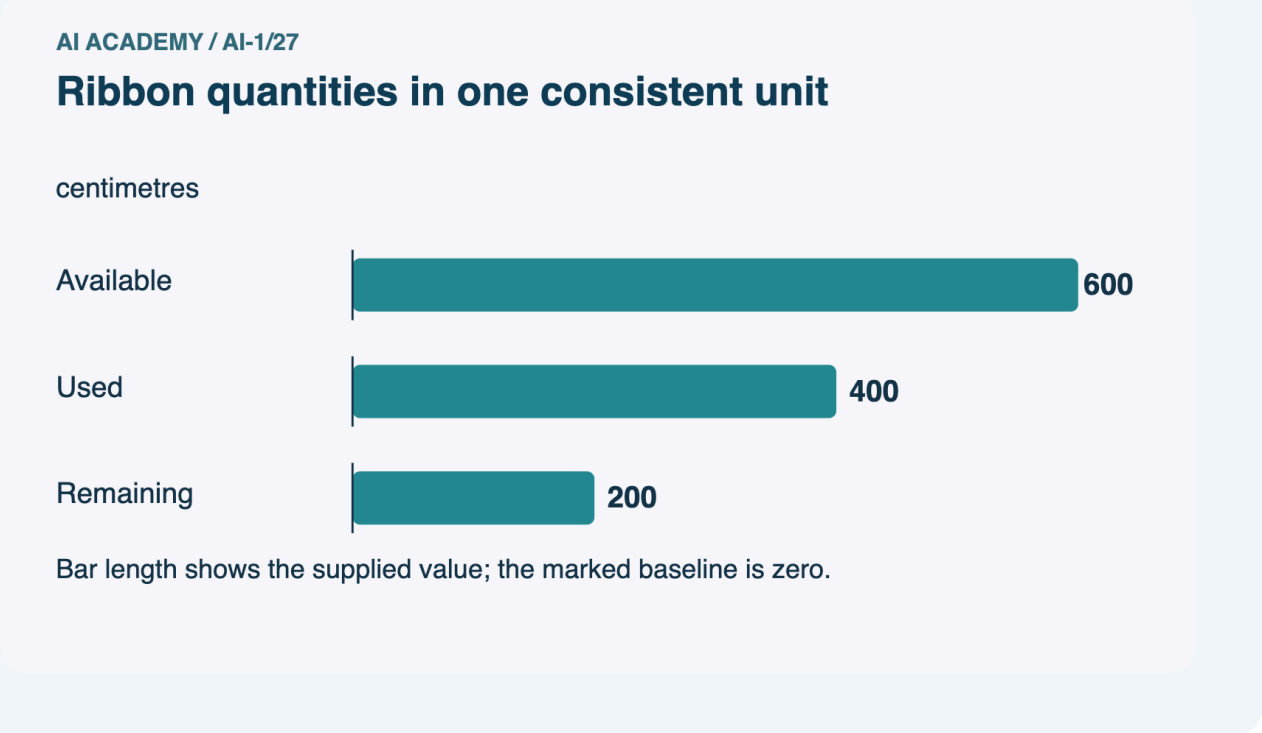


Follow the labelled picture with the matching explanation. Open the vector diagram

Available ribbon: 4×150=600 cm. → Ribbon used: 8×50=400 cm. → Remainder: 600−400=200 cm=2 m. → Audit draft: “200 m remains and everyone wins” fails **unit** and **evidence** checks.

Point to each numbered stage and explain which recorded **input** it uses.

03. Ribbon quantities in one consistent unit

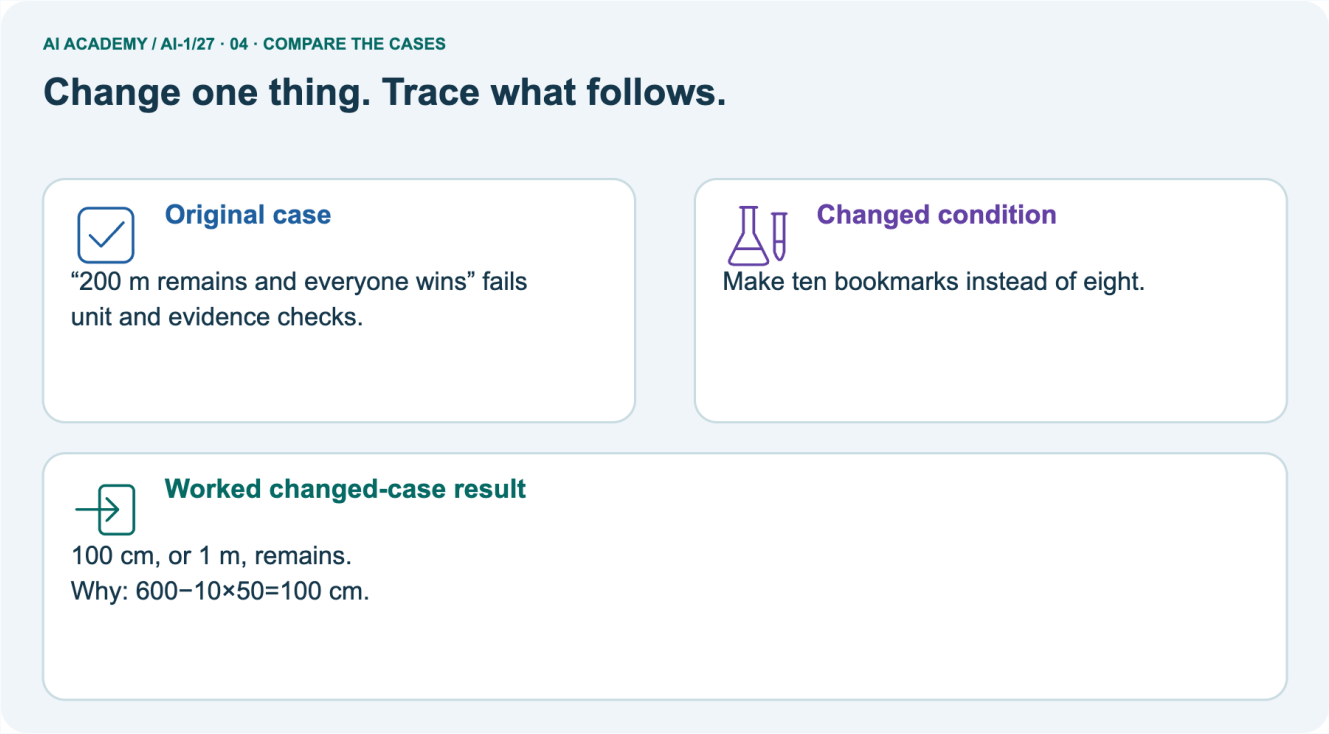


Follow the labelled picture with the matching explanation. Open the vector diagram

Ribbon quantities in one consistent **unit**. **Unit**s: centimetres. Available: 600. Used: 400. Remaining: 200.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.

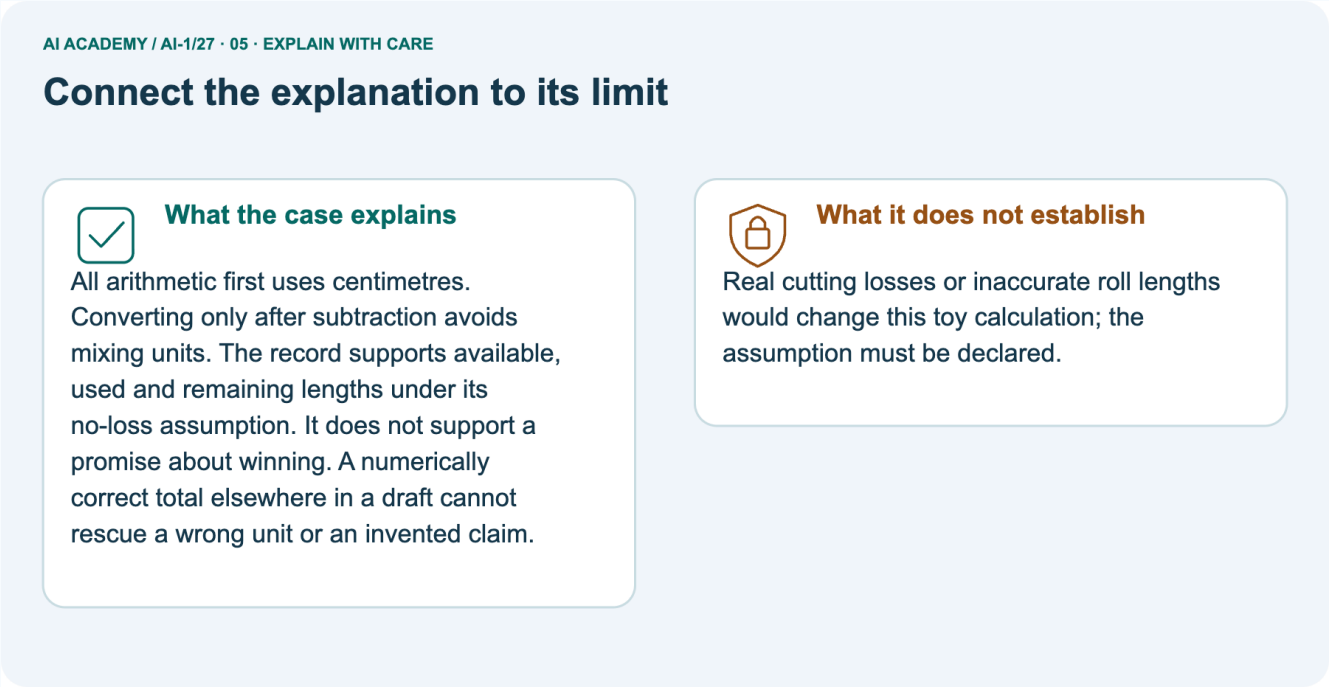


Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: “200 m remains and everyone wins” fails **unit** and **evidence** checks. Changed condition: Make ten bookmarks instead of eight. Expected result: 100 cm, or 1 m, remains. Reason: 600−10×50=100 cm.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



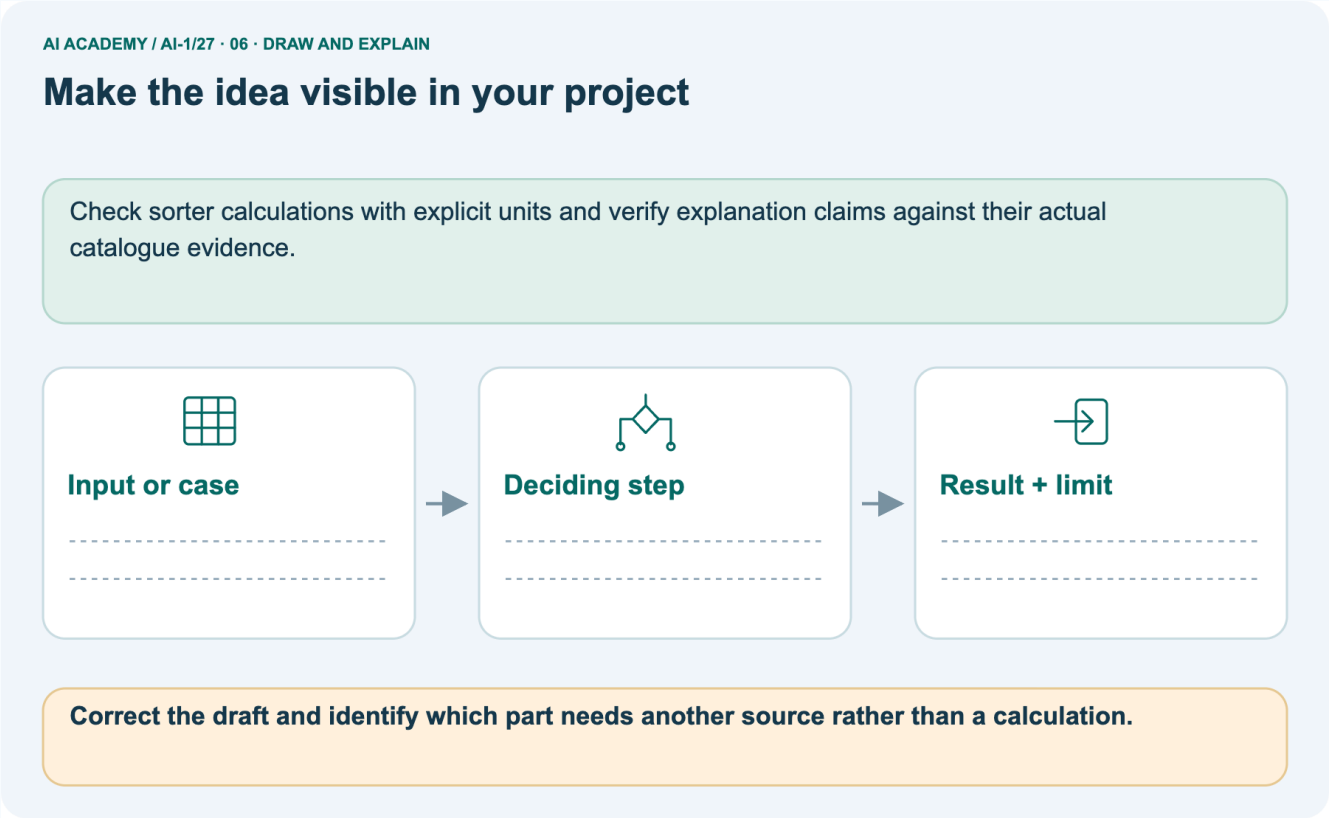
Follow the labelled picture with the matching explanation. Open the vector diagram

All arithmetic first uses centimetres. Converting only after subtraction avoids mixing **unit**s. The record supports available, used and remaining lengths under its no-loss assumption. It does not support a promise about winning.

A numerically correct total elsewhere in a draft cannot rescue a wrong **unit** or an invented **claim**. Limit: Real cutting losses or inaccurate roll lengths would change this toy calculation; the assumption must be declared.

Explain the case, then point to a limit on the **claim**.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Check sorter calculations with explicit **unit**s and verify explanation **claim**s against their actual catalogue **evidence**. Draw your **input**, deciding step, result and limit.

Supported question: Correct the draft and identify which part needs another **source** rather than a calculation.

Draw your project trace and justify the deciding step using **evidence**.

Earlier wording, alternatives and picture transcripts

01. Verify Important Outputs: the case

02. Follow the deciding steps

03. Ribbon quantities in one consistent unit

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.